

04-July-2026

Agenda:

ML-14

→ Support Vector Machine (SVM)

→ primal form (linear SVM) → completed

→ Dual form SVM

→ Regression → SVR

→ practicals (SVM + gridsearchcv)

$$J(w, b) = \frac{1}{2} \|w\|^2 + c \sum_{i=1}^n \max(0, 1 - y_i (w \cdot x_i + b))$$

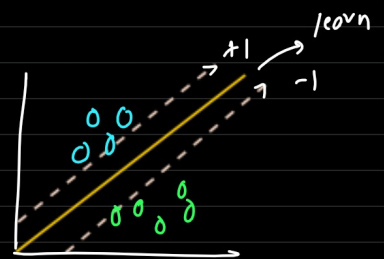
path A: ($z \geq 1$), loss $\rightarrow 0$

$$\frac{\partial J}{\partial w} = w \quad \frac{\partial J}{\partial b} = 0$$

path B: ($z < 1$), loss is active

$$\frac{\partial J}{\partial w} = w - (c \cdot y_i \cdot x_i)$$

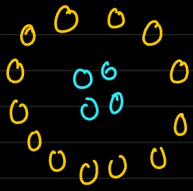
$$\frac{\partial J}{\partial b} = -(c \cdot y_i)$$



The problem: when straight line fails

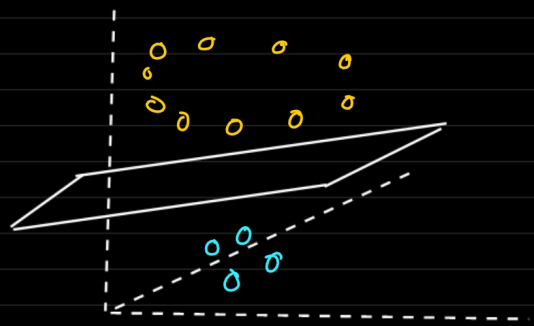
linear regression $\rightarrow$ polynomial regression





2D

transform the data
into higher dimension



(Dual form - kernelized SVM)

$$A: [2, 3] \rightarrow$$

2D

(polynomial function)

$$\phi(x) =$$

$d=2$

$$[x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

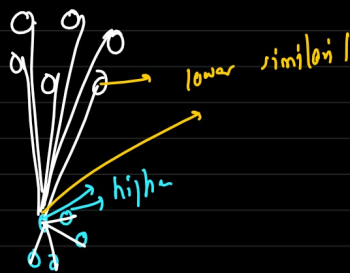
$$A: [2^2, \sqrt{2(2)(3)}, 3^2]$$

$$[4, 6\sqrt{2}, 9]$$

3D

$$B: [4, 1]$$

$$[16, 4\sqrt{2}, 1]$$



Requirement:

1) mathematical function $\rightarrow$ high dimensional data

2) close or similar to their class $\rightarrow$ dot product

$$A: [4, 6\sqrt{2}, 9] \quad B: [16, 4\sqrt{2}, 1]$$

→ Take the 3D dot product:

$$(4 \times 16) + (6\sqrt{2} \times 4\sqrt{2}) + (9 \times 1) = 64 + (24 \times 2) + 9 \\ = 121$$

2D → 3D → dot product → single value ← similarity score

A-B

closer → higher score

far → lower score

2D → 3D

100,000

3D

□ □ □

→ memory
→ computation

100,000 × 3

computation

dot product score

$$\rightarrow (A \cdot B)^2$$

step-1 → A → 3D

step-2 → B → 3D

step-3 → A dot B

$$\left(\begin{bmatrix} 2, 3 \end{bmatrix} \cdot \begin{bmatrix} 1, 1 \end{bmatrix} \right)^2 \rightarrow \left((2 \times 1) + (3 \times 1) \right)^2 \rightarrow \left((2) + (3) \right)^2 \\ \rightarrow (5)^2 \rightarrow 25$$

kernel function

or

kernel trick

$$A - B \rightarrow \square$$

$$A - C \rightarrow \square$$

computer

3D

$$\text{car} \rightarrow (\text{red}, 240 \text{ km/hr}, 40) \rightarrow A$$

$$\text{car} \rightarrow (\text{yellow}, 120 \text{ km/hr}, 20) \rightarrow B$$

$$\text{car} \rightarrow (\text{yellow}, 230 \text{ km/hr}, 35) \rightarrow C$$

$$\text{car} \rightarrow (\text{blue}, 110 \text{ km/hr}, 5) \rightarrow D$$

$$\text{car} \rightarrow (\text{green}, 130 \text{ km/hr}, 10) \rightarrow \square$$

3D

B, C

2D $\rightarrow$ ND

linear kernel:

SVM $\rightarrow$ kernel $\rightarrow$ linear

$$\phi(x) = x$$

$$\phi(2,3) = [2, 3]$$

$$\phi(2,3)$$

$$\phi(4,1)$$

A · B

polynomial kernel:

$$\phi(x) : [x_1^2, \sqrt{2}x_1x_2, x_2^2]$$

$d=2$

$$\phi(2,3) : [4, 6\sqrt{2}, 9] \rightarrow (A \cdot B)^2$$

$$\phi(4,1) : [16, 4\sqrt{2}, 1]$$

RBF kernel (Radial Basis function / Gaussian)

maps data to infinite dimension

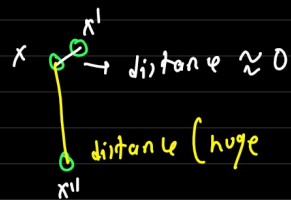
$\rightarrow$ Taylor series $\rightarrow e^x \rightarrow$ polynomial goes forever

$$k(x, x') = \exp(-\gamma \|x - x'\|^2)$$

$\rightarrow$ square euclidean distance between two point

data $\rightarrow$ infinite $\rightarrow$ kernel $\rightarrow$

instead of thinking about infinite dimension, think about close & far points are



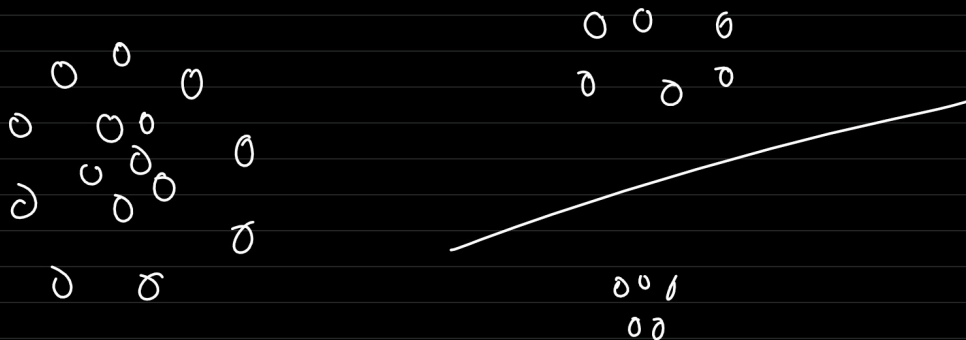
point right on top of each other
 $\rightarrow$ distance 0

$\rightarrow \exp(0) \rightarrow 1$

point that are miles apart
 $\rightarrow$ distance huge

$\rightarrow \exp(-\text{huge}) \approx 0$

RBF $\rightarrow$ similarity score ranging from 1 (identical) to 0 (unrelated)



→ 10 → 20 → 5 → 4
 Hello, How are you $\rightarrow$ ① $[10, 20, 5, 4]$

Hello, are you How $\rightarrow$ ② $[10, 5, 4, 20]$

Hello $\rightarrow$ $[\text{D D D D D D D D }]$
 Hello, $\rightarrow$ $[\text{D D D D D D D D }]$
 (Note: The second row has a "1x512" label above the closing bracket and a "1x512" label below the closing bracket.)

SVM:

we want to minimize, subject to

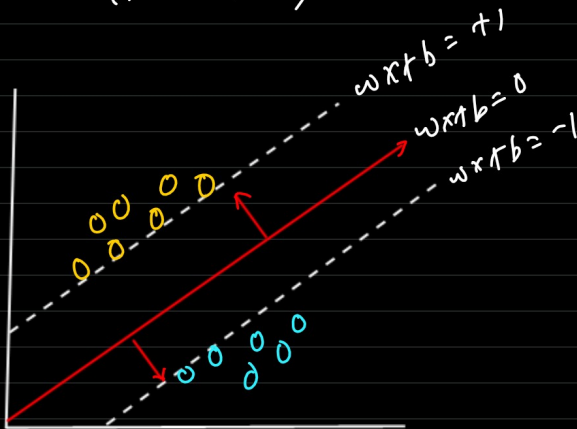
$$\frac{1}{2} \|w\|^2$$

$$y_i (w \cdot x_i + b) \geq 1$$

$$Z \geq 1 \rightarrow \text{no loss}$$

$$Z < 1 \rightarrow \text{learning}$$

SVM (primal form)



learnable parameters: w, b

SVM (dual form)



idea:

→ increase the dimension of the data to make them separable.

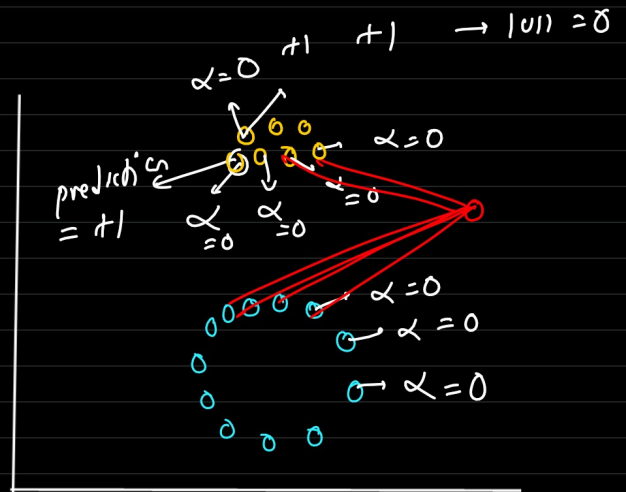
↓
2D → 3D
 $x_1, x_2 \quad x_1, x_2, x_3$

↓ 2D → 4D
↓ 2D → 5D } issue: computation

formula +
dot-product → single score

↓
kernel

student → Age, class, height, score(A, B, C, D, E)
(A)(8) →
(B)(8) →



→ Dual form, the kernel is strictly a pair-wise operation

prediction $\leftarrow$ formula [voting from all pairs]
 $= +1$

Actual $\rightarrow$ (OS) $\rightarrow$

10,000 data $\rightarrow$ learnable param $\rightarrow \alpha_{total} \rightarrow 10,000$

$\rightarrow \alpha_{total} \rightarrow 50 - 100$

→ kernel ✓

→ primal & dual form

↓
 → kernel
 → similarity concept

$2(w)$	$3(w)$	RBF
$x_1 \ x_2$	$x_1 \ x_2 \ x_3$	$3(\infty)$
2D $\rightarrow$	3D	2D $\rightarrow$ ND

prediction $\rightarrow y_i(\omega \cdot x + b)$ (primal form)

prediction $\rightarrow \sum_{i=1}^n \alpha_i y_i (k(x_i, x) + b)$ (dual form)

Iteration 1: $d=2, b=0, \alpha=0.01$

$$k(x, x') = (x \cdot x')^2$$

precomputed similarities

→ select a data point

→ A

	x_1	x_2	y_A
A	2	3	+1
B	4	1	-1
C	0	1	+1

 → training loop 1:

$$\alpha_A \alpha_B \alpha_C = 0$$

$$k(A, A) = ((2 \times 2) + (3 \times 3))^2 = 13^2 = 169$$

$$k(B, A) = ((4 \times 2) + (1 \times 3))^2 = 11^2 = 121$$

$$k(C, A) = ((0 \times 2) + (1 \times 3))^2 = 3^2 = 9$$

→ step-2 (prediction A) $\sum_{i=1}^n \alpha_i y_i (k(x_i, x) + b)$

$$\text{vote from A: } \alpha_A \cdot y_A \cdot k(A, A) = 0.1 \cdot 169 = 0$$

$$\text{vote from B: } \alpha_B \cdot y_B \cdot k(B, A) = 0 \cdot -1 \cdot 121 = 0$$

$$\text{vote from C: } \alpha_C \cdot y_C \cdot k(C, A) = 0 \cdot 1 \cdot 9 = 0$$

$$\text{Total-prediction for point A} = \text{vote}(A + B + C) = 0.0$$

step-3 → Check Error (Hinge loss)

$$z = y_A \cdot \text{prediction}$$

$$z = (1) \cdot (0) = 0.0$$

$z < 1 \leftarrow$ Error, update learning rate

$$\max(0, 1 - z) \quad \text{primal form}$$

$$\max(0, 1 - y_i (w \cdot x_i + b))$$

$$\max(0, 1 - y_i (\alpha_i y_i (k(x_i, x) + b)))$$

$$\max(0, 1 - z) \quad \text{dual form}$$

step-4: update learning rate

$$\alpha_A = \alpha_A + \text{learning rate}$$

$$= 0 + 0.01 = 0.01$$

Step-5 : Re-predict point A

$$\text{vote-A} : \alpha_A \cdot y_A \cdot k(A, A) = 0.01 \cdot 1 \cdot 1.69 = 1.69$$

$$\text{vote-B} : 0$$

$$\text{vote-C} : 0 \\ = 1.69$$

$$z = y_i \cdot \text{prediction} \rightarrow (1)(1.69) \rightarrow 1.69$$

$z > 1 \rightarrow$ point is in safe zone

Iteration: 2

Step-1 compute similarity

	x_1	x_2	y_A	α
A	2	3	+1	0.01
B	4	1	-1	0
C	0	1	+1	0

$\rightarrow$ training

$$k(A, B) = 121$$

$$k(B, B) = ((4 \times 4) + (1 \times 1))^2 = 289$$

$$k(C, B) = ((0 \times 4) + (1 \times 1))^2 = 1$$

Step-2 : prediction B

$$\text{vote from A} : \alpha_A \cdot y_A \cdot k(A, B) = 0.01 \cdot (1) \cdot 121 = 1.21$$

$$\text{vote from B} : \alpha_B \cdot y_B \cdot k(B, B) = 0 \cdot (-1) \cdot 289 = 0$$

$$\text{vote from C} : \alpha_C \cdot y_C \cdot k(C, B) = 0 \cdot (1) \cdot 1 = 0$$

$$\text{Total predicted} = \text{vote (A+B+C)} \\ = 1.21 \rightarrow$$

vote from A pulled is
telling that A & B are
similar

Step:3 $\rightarrow$ Error

$$z = y_B \cdot \text{prediction} \rightarrow (-1)(1.21) = -1.21$$

$-1.21 > 1 \rightarrow$ Massive Error

step: 4 learning

$$\alpha_\beta = \alpha_\beta + \text{learning} \\ = 0.0 + 0.01 \rightarrow 0.01$$

step: 5 evaluate

$$\text{vote-A} : \alpha_A \cdot y_A \cdot k(A, B) = 0.01 \cdot 1 \cdot |21| = 1.21$$

$$\text{vote-B} : 0.01 \cdot (-1) \cdot 289 = -2.89$$

$$\text{vote-C} : 0$$

$$= 1.21 - 2.89 + 0 \rightarrow -1.68$$

$$z = y_i \cdot \text{prediction} \rightarrow (-1)(-1.68) \rightarrow 1.68$$

$z > 1 \rightarrow$ point is in safe zone

$$\text{previous } z \rightarrow -2.89$$

$$\text{new } z \rightarrow 1.68 > 1 \rightarrow \text{safe}$$

primal form

learnable
params

weight (w) and bias (b)

prediction

$$\text{prediction} = w \cdot x + b$$

confidence

$$z = y \cdot \text{prediction}$$

learning
flag

Is $z < 1$?
 $\rightarrow$ update

Dual form

temporale score (α) for each point
& bias (b)

$$\text{prediction} = \sum \alpha_i y_i k(x_i, x) + b$$

$$z = y \cdot \text{prediction}$$

Is $z < 1$?
 $\rightarrow$ update

update
Math

$$w_{\text{new}} = w_{\text{old}} - \alpha (\text{gradient})$$

$$\alpha_{\text{new}} = \alpha_{\text{old}} \times \text{learning rate}$$

mental
model

tilt & shift a rigid 2D
highway (centre line &
margin)

Discovering boundary points via weighted
similarity search